$$g(y)=(y-4)(2y+y^2)$$
$$g(y)=2y^2+2y^3-8y-4y^2$$
$$\frac{dg}{dy}=4y+6y^2-8-8y$$
$$\frac{dg}{dy}=6y^2-8y+4y-8$$
$$\frac{dg}{dy}=6y^2-4y-8$$